Ken Shoulders' EVs

Charge clusters of 10^{11} electrons checked against Coulomb, Gauss, the Casimir force and charge and energy budgets — register entry H27

kronspice simulation report · 2026-11-07 · bench values declared

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Summary

K. R. Shoulders (*EV: A Tale of Discovery*, 1987) reports micrometre beads of electric charge thrown off by fast discharges from sharp cathodes, which he calls EVs (*electrum validum*). He reports about 10^{11} electrons per 1 μm bead, near-solid electron density with almost no ions, motion at about c/10, and he proposes that the zero-point fluctuations of empty space hold them together. This example takes his numbers at face value and checks them against Coulomb's and Gauss's laws, the known containers, and the charge and energy budgets of his sources; then it runs his source and an EV strike as kronspice circuits with energy ledgers. Page numbers below are the book's chapter-page numbers.

- **Energy:** a 1 μm bead of 10^{11} electrons needs **2.77 J** to assemble, 3 million times his smallest source's 1 μJ and 1,400 times his larger source's 2.0 mJ.
- **Left bare it explodes:** outward pressure 1.5×10^{18} Pa; the radius doubles in 1.7×10^{-15} s, 20,000 times shorter than his reported lifetime of 3.3×10^{-11} s.
- **No known container:** the Casimir pull (1.3×10^{-3} Pa) is 10^{21} times too weak, and on a sphere the zero-point stress pushes outward; a magnetic trap would need 2×10^{5} T.
- **Gauss's law:** his kilovolt readings need a “container” of at least 14.4 cm radius (1 kV), or a net charge of at most about 3×10^{5} electrons.
- **Charge budget:** the 1 pF source holds 8.8×10^{9} electrons; one EV bead is 11 times that. In kronspice the source puts exactly C·V (8.83×10^{9} electrons) through the scope, ledger closed to 3×10^{-5}.
- **Test:** a one-shot Faraday-cup charge measurement against C·V of the source. Standard physics predicts it never exceeds C·V.

The claim and the standard prediction

Hypothesis (H27), from Shoulders 1987:

- **Size and charge:** about 1 μm across, often in chains and rings; one bead carries “around 10^{11} charges” (8-49).
- **Content:** electrons at a density near that of a solid (6.6×10^{23} per cm³), with fewer than 1 ion per 100,000 electrons (4-63).
- **Motion and life:** about c/10, living about 10^{-11} s; they bore pits, leave ring-shaped marks and give X-rays where they hit.
- **Binding:** the zero-point fluctuations of empty space (Preface; 9-2).
- **Source:** a cathode of about 1 pF storing less than 10^{-6} J (2-7).

Standard prediction. A bead of 10^{11} electrons in 1 μm is held by nothing known and explodes in femtoseconds; the charge collected from one shot can never exceed C·V of the source, and the energy delivered never exceeds what the source stored. The measured counterpart is G. A. Mesyats' ectons: explosive-emission micro-explosions at cathode spots.

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1. A bead of 10^{11} electrons, 1 μm across

quantity	value
charge	1.6×10^{-8} C
density	1.9×10^{23} electrons/cm³ (his 3.5×10^{14} electrons in 10 μm gives his 6.6×10^{23})
energy needed to assemble it	**2.77 J**
field at its surface	5.8×10^{14} V/m
outward pressure	1.5×10^{18} Pa (10^{13} atmospheres)
kinetic energy of all its electrons at c/10	4.1×10^{-5} J (the field holds 67,000 times more)

Left bare, it flies apart. The outer electrons are pushed off at nearly the speed of light: the radius doubles in 1.7×10^{-15} s and grows tenfold in 1.5×10^{-14} s, and each outer electron leaves with about 260 MeV. Shoulders' lifetime (1 mm of track at c/10) is 3.3×10^{-11} s, 20,000 times longer than the doubling time. Something would have to hold it.

2. What could hold it?

candidate	pressure	against the 1.5×10^{18} Pa push
magnetic attraction of electrons moving together at c/10	1 % of the repulsion	the push still wins
a magnetic trap at this density (Brillouin limit)	needs 2×10^{5} T	the strongest field made indoors is about 1,200 T
Casimir pull between ideal plates 1 μm apart	1.3×10^{-3} Pa	10^{21} times too weak
zero-point stress on a 1 μm conducting sphere	1.9×10^{-3} Pa, **outward**	pushes the wrong way (Boyer 1968)
positive ions	would cancel the charge	his own measurement: under 1 ion per 100,000 electrons

The zero-point field is the container Shoulders proposes. Its known, measured effect, the Casimir force, is 21 orders of magnitude too weak, and for a sphere it pushes outward.

3. Gauss's law and his kilovolt measurements

Shoulders states the puzzle himself (9-2): an electron leaving a bare 1 μm bead of 10^{11} electrons gains 288 MeV, but his probes saw “only kilovolts at best”. His answer is a container that hides the core's field. Gauss's law rules this out: outside any closed surface, the potential of a net charge q is k·q/r whatever is inside. For escaping electrons to gain only V volts the “container” must be at least:

escape energy seen	minimum container radius
1 kV	14.4 cm
10 kV	1.4 cm

Those sizes are larger than the millimetre gaps of his apparatus. The alternative is a net charge of at most about 3×10^{5} electrons for a 1 μm bead seen at 1 kV, 300,000 times less than 10^{11}, unless positive charge cancels it, which his ion measurement excludes.

4. Charge and energy budgets

source (2-7)	electrons it holds (C·V)	energy
smallest: 1 pF, < 1 μJ (at most 1.4 kV)	8.8×10^{9}	1.0 μJ
earlier: 40 pF at 10 kV	2.5×10^{12}	2.0 mJ

what he reports	electrons	÷ small source	÷ larger source
residual charge of a 3 μm EV on an electrode (4-63)	2×10^{10}	2.3	0.008
one EV bead (8-49)	10^{11}	11	0.04
emitted by a 10 μm EV over 1 mm of track (4-9)	3.5×10^{14}	40,000	**140**

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- **Charge** is conserved exactly. A 10^{11}-electron bead cannot come from the 1 pF source, and the 3.5×10^{14} electrons he infers from camera brightness cannot come from either source. That count is probably the inference to check first: it rests on a phosphor calibration and on isotropic emission.
- **Energy:** the 2.77 J assembly energy is 3 million times the small source's energy and 1,400 times the larger one's.

5. kronspice: the source and a strike, with ledgers

The bench (declared values): a 1 pF anode, the witness plate or a collector, feeding a 50 Ω scope line.

a. His smallest source

A 1 pF cathode at 1.414 kV (1 μJ) discharges across the gap (a 20 Ω arc with 5 nH).

quantity	value
charge through the scope	8.83×10^{9} electrons (all of C·V)
energy	1.000 μJ stored → 0.638 μJ to the scope, 0.362 μJ heat in the arc
ledger	closes to 3×10^{-5}

b. An EV strike

An ideal current source brings the EV's charge to the anode in 20 ps.

electrons	peak at the scope	energy the arriving charge must bring	÷ source energy
2×10^{10}	2.7 kV	4.7 μJ	5
10^{11}	13.4 kV	117 μJ	117

The ledger closes to 10^{-5} in both.

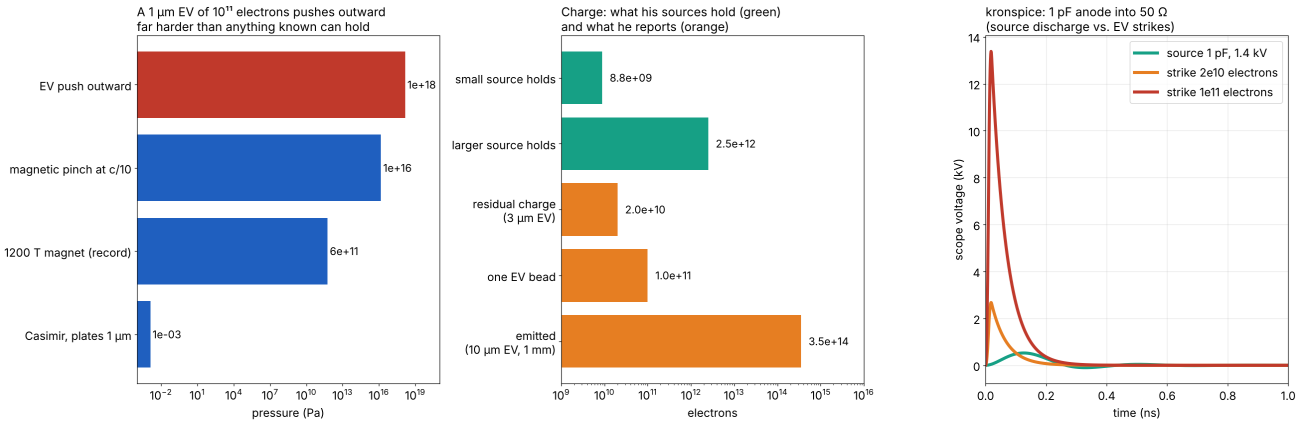


Figure 1. Left: outward pressure of a 1 μm EV of 10^{11} electrons against the magnetic pinch at c/10, the pressure of the record 1200 T magnet and the Casimir pull between plates 1 μm apart (log scale). Middle: electrons his sources hold (green) against what he reports (orange). Right: kronspice scope voltage on the 1 pF anode into 50 Ω for the 1 pF, 1.4 kV source discharge and for EV strikes of 2×10^{10} and 10^{11} electrons.

6. The rings, and the mainstream counterpart

- **Ring marks.** Ring-shaped witness marks (2-19) fit a toroidal current ring. A real electron ring needs an outside field to hold it: the collapsing ring in *examples/plasmoid_rings* needs a 2 T coil, and its electrons are 10^{6} times sparser than an EV's.
- **Ectons.** Mesyats' ectons are explosive-emission micro-explosions at cathode spots, each lasting nanoseconds and releasing about 10^{11}–10^{12} electrons. They leave craters and fast current pulses and give X-rays.
- **The difference.** An ecton is a neutral plasma of cathode metal, with ions. Shoulders reports almost no ions, so his ion measurement is where the two pictures disagree. It is worth repeating.

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Validation

- *main_test.go* checks the electrostatics, the Coulomb explosion against the exact solution, the containment numbers, and the kronspice charge and ledger.
- kronspice ledgers close to 3×10^{-5} (source discharge) and 10^{-5} (strikes); the source delivers exactly its C·V.

Limits of the model

- Shoulders' reported numbers are taken at face value; the bench (1 pF anode, 50 Ω scope, 20 Ω arc with 5 nH) is declared.
- The EV strike is an ideal current source delivering the charge in 20 ps; it shows the energy the charge must bring, not a model of how it is made.
- The observations themselves (pits, rings, chains, X-rays, very fast pulses from tiny discharges) are real and carefully documented; what is tested here is the EV interpretation.

Status

At the reported size and charge an EV holds 1,400 to 3 million times its source's energy, flies apart in femtoseconds, shows a field at odds with Gauss's law and his own kilovolt measurements, and has no known container. What would change this: a one-shot Faraday-cup charge measurement exceeding C·V of the source, with this ledger on the shot. That is the H27 test. Shoulders' later patents (5,018,180; 5,054,046; 5,054,047; 5,123,039; 5,148,461) and the Puthoff & Piestrup review carry the same claims.

Reproducing the results

```
go run ./examples/shoulders_ev -viz examples/shoulders_ev/viz   # under a second
python3 examples/shoulders_ev/viz/render.py
go test ./examples/shoulders_ev
```

Sources: K. R. Shoulders, *EV: A Tale of Discovery* (1987); H. E. Puthoff, M. A. Piestrup, arXiv:physics/0408114 (2004); T. H. Boyer, Phys. Rev. 174, 1764 (1968); H. B. G. Casimir, Proc. K. Ned. Akad. Wet. 51, 793 (1948); L. Brillouin, Phys. Rev. 67, 260 (1945); G. A. Mesyats, Plasma Phys. Control. Fusion 47, A109 (2005); D. Nakamura et al., Rev. Sci. Instrum. 89, 095106 (2018).

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